Execution Environment
(Dual-pipeline UE + Edge queues)

UE Devices

Uplink radio channel

Edge Servers (1..M)

Edge-Cloud wired connection

Cloud Center

Local CPU channel

Local Execution (on-device)

s_{t+1}

Task selection

(A) Online Observations

$s_t = (G_t, \text{queues}, \text{pipelines})$

Task DAG snapshot

red: rank_u-based

arrivals

(B) Ready Queue (slot t)

T1 T5 T7

T9 T3 T2

task list

(C) CP Sequencer (deterministic)

$\text{Pri} = (\text{slack}_{cp}, -\text{rank}_u)$

T7 T1 T9

T5 T2 T3

a_t

selected task

Placement decision

(D) Dual-GAT

Forward : dependency propagation

Fusion

Backward : successor deadline pressure

Context

(E) Feasible Mask

Local ✓ Edge1 ✗

Edge2 ✗ Edge3 ✓

...

Valid actions

(F) Deadline-Risk Shield

within budget √ → retain

exceeds budget X → filter

safe actions

(G) Masked Actor

Local ✓ Edge1 ✗ Edge2 ✗ Edge3 ✓ ... Cloud ✓

Placement a_t